

The conjugation action of $\mathrm{SL}_2(\mathbb{R})$ is Rosenthal representable

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Status

Full candidate solution. This packet gives a positive answer to Question 3.20 of Megrelishvili's paper *Topological group actions by group automorphisms and Banach representations*, arXiv:2110.01386.

in contrast, the left action of $\mathcal{U}_\infty$ on itself is always unitarily representable and the topological group S_∞ is unitarily representable on a Hilbert space.

Question 3.20. *Is it true that for $G = \mathrm{SL}_2(\mathbb{R})$ the G -space G_c is Rosenthal representable?*

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Answer

Yes. For $G = \mathrm{SL}_2(\mathbb{R})$, the conjugation G -space G_c is Rosenthal representable.

The proof below gives a slightly stronger matrix-group statement. The extra generality is included because it makes the mechanism transparent: conjugation is a finite-dimensional linear action once the group is viewed inside a matrix algebra.

Proof Intuition

The source paper proves that finite-dimensional linear actions are tame after one-point compactification: the enveloping semigroup consists of partial linear maps, so it has cardinal at most continuum. Now $\mathrm{SL}_2(\mathbb{R})$ sits as a closed subset of the four-dimensional vector space $M_2(\mathbb{R})$, and conjugation by $g \in \mathrm{SL}_2(\mathbb{R})$ is the linear map $A \mapsto gAg^{-1}$ on that vector space.

Therefore the one-point compactification of $M_2(\mathbb{R})$ is a tame compact $\mathrm{SL}_2(\mathbb{R})$ -space. The subset $\mathrm{SL}_2(\mathbb{R}) \cup \{\infty\}$ is closed and invariant inside it; closed invariant subsystems of tame compact systems are tame. This gives a tame proper compactification of G_c , which is exactly what is needed for Rosenthal representability.

Definitions used

For a compact metric G -space X , the source paper recalls the standard equivalence: X is representable on a Rosenthal Banach space if and only if X is tame; equivalently, its enveloping semigroup $E(X)$ has cardinal at most $2^{\aleph_0}$ [1, Fact 3.11].

A G -space X is Rosenthal representable if it admits a proper G -compactification by a compact metric tame G -space, using the source paper's representation definition and compact-metric tame characterization.

The finite-dimensional lemma

Lemma 1. *Let V be a finite-dimensional real vector space and let $\alpha V = V \cup \{\infty\}$ be its Alexandrov one-point compactification. The natural action $\mathrm{GL}(V) \curvearrowright \alpha V$ is tame.*

Proof. Let $p \in E(\alpha V)$. Choose a net $T_i \in \mathrm{GL}(V)$ converging pointwise to p on αV . Define

$$D_p = \{v \in V : p(v) \in V\}.$$

If $u, v \in D_p$ and $a, b \in \mathbb{R}$, then $T_i u \rightarrow p(u)$ and $T_i v \rightarrow p(v)$ in V , hence

$$T_i(au + bv) = aT_i u + bT_i v \rightarrow ap(u) + bp(v) \in V.$$

Thus $au + bv \in D_p$ and $p|_{D_p}$ is linear. Since V is finite dimensional, D_p is a closed linear subspace and $p|_{D_p} : D_p \rightarrow V$ is continuous. For $v \notin D_p$, by definition $p(v) = \infty$.

Consequently each element of $E(\alpha V)$ is determined by a linear subspace $D \leq V$ and a linear map $D \rightarrow V$. There are only $2^{\aleph_0}$ such pairs. Hence $\mathrm{card} E(\alpha V) \leq 2^{\aleph_0}$, and by the source paper's tameness criterion the compact system αV is tame. $\square$

Lemma 2. *Let X be a compact tame G -space and let $Y \subset X$ be a closed G -invariant subset. Then Y is tame.*

Proof. For every $p \in E(X)$, $p(Y) \subset Y$: indeed p is a pointwise limit of translations by elements of G , each of which sends Y to itself, and Y is closed. Restriction gives a map

$$E(X) \longrightarrow Y^Y, \quad p \mapsto p|_Y.$$

Every element of $E(Y)$ is obtained this way after passing to a convergent subnet in the compact space X^X . Therefore $\mathrm{card} E(Y) \leq \mathrm{card} E(X) \leq 2^{\aleph_0}$, so Y is tame. $\square$

Main theorem

Theorem 1. *Let $G \leq \mathrm{GL}_m(\mathbb{R})$ be a matrix group with the topology inherited from $\mathrm{GL}_m(\mathbb{R})$. Then the conjugation G -space G_c is Rosenthal representable.*

Proof. Put $V = M_m(\mathbb{R})$. The formula

$$g \cdot A = gAg^{-1} \quad (g \in G, A \in V)$$

defines a continuous finite-dimensional linear action of G on V , because $A \mapsto gAg^{-1}$ is a linear automorphism of the vector space V .

Let $\alpha V = V \cup \{\infty\}$ be the one-point compactification and let

$$K = \overline{G}^{\alpha V}$$

be the closure of the image of $G \subset V$. The inclusion $G \hookrightarrow K$ is a topological embedding with dense image, and the compact space K is G -invariant under the extended linear action. On the dense copy of G , this action is exactly conjugation:

$$g \cdot x = gxg^{-1}.$$

By the first lemma, αV is tame for the full linear group $\mathrm{GL}(V)$, hence also for the restricted G -action. By the second lemma, the closed invariant subsystem $K \subset \alpha V$ is tame. Thus K is a tame compact metric G -compactification of G_c . By the compact-metric tame/Rosenthal equivalence from the source paper, G_c is Rosenthal representable. $\square$

Corollary 1. *For $G = \mathrm{SL}_2(\mathbb{R})$, the conjugation G -space G_c is Rosenthal representable. Hence Question 3.20 of [1] has a positive answer.*

Proof. $\mathrm{SL}_2(\mathbb{R})$ is a matrix group in $\mathrm{GL}_2(\mathbb{R})$ with its usual topology, so the theorem applies. In this case $\mathrm{SL}_2(\mathbb{R}) = \{A \in \mathrm{M}_2(\mathbb{R}) : \det A = 1\}$ is closed in $\mathrm{M}_2(\mathbb{R})$, so the compactification K above is simply the one-point compactification $\mathrm{SL}_2(\mathbb{R}) \cup \{\infty\}$. $\square$

Corollary 2. *For every $n \geq 2$, the conjugation action of $\mathrm{SL}_n(\mathbb{R})$ is Rosenthal representable.*

Proof. Apply the theorem to the usual matrix embedding $\mathrm{SL}_n(\mathbb{R}) \leq \mathrm{GL}_n(\mathbb{R})$. $\square$

Verification notes

The proof avoids the common false step “a tame orbit implies the whole action is tame.” Instead it embeds the entire conjugation G -space in a compact tame linear system. For $G = \mathrm{SL}_2(\mathbb{R})$, no singular matrix boundary is added, since determinant one is closed in $\mathrm{M}_2(\mathbb{R})$. For nonclosed or open matrix groups such as $\mathrm{GL}_m(\mathbb{R})$, the same theorem uses the closure in $\alpha \mathrm{M}_m(\mathbb{R})$; this is still a valid compactification because the original group embeds densely and equivariantly.

This full packet supersedes the earlier partial packet proving only that the unipotent conjugacy orbit is Rosenthal representable.

Novelty and literature check

Local run indexes were searched for arXiv id 2110.01386, the title, $\mathrm{SL}_2(\mathbb{R})$, G_c , conjugation, matrix groups, finite-dimensional linear action, and Rosenthal representability. The only matching durable packet was the earlier partial unipotent-orbit packet from this lane.

Web searches on June 18, 2026 for exact phrases including $\mathrm{SL}_2(\mathbb{R})$, G_c , conjugation action, Rosenthal representable, matrix group, and finite-dimensional linear action found the source paper but no later paper explicitly resolving Question 3.20. The arXiv abstract for the source paper itself states that the $\mathrm{SL}_2(\mathbb{R})$ case is unclear [2]. Thus this is recorded as a full candidate solution rather than a literature-status identification.

Human review recommendation

Review the two heredity points: first, that the closure of G inside $\alpha \mathrm{M}_m(\mathbb{R})$ is a legitimate G -compactification of the conjugation G -space; second, that tameness descends to closed invariant subsystems through restriction of enveloping semigroups. If both are accepted, the proof gives a positive solution of Question 3.20.

References

- [1] M. Megrelishvili, *Topological group actions by group automorphisms and Banach representations*, Forum Mathematicum, accepted version; arXiv:2110.01386, 2023.
- [2] arXiv page for M. Megrelishvili, *Topological group actions by group automorphisms and Banach representations*, <https://arxiv.org/abs/2110.01386>, accessed June 18, 2026.